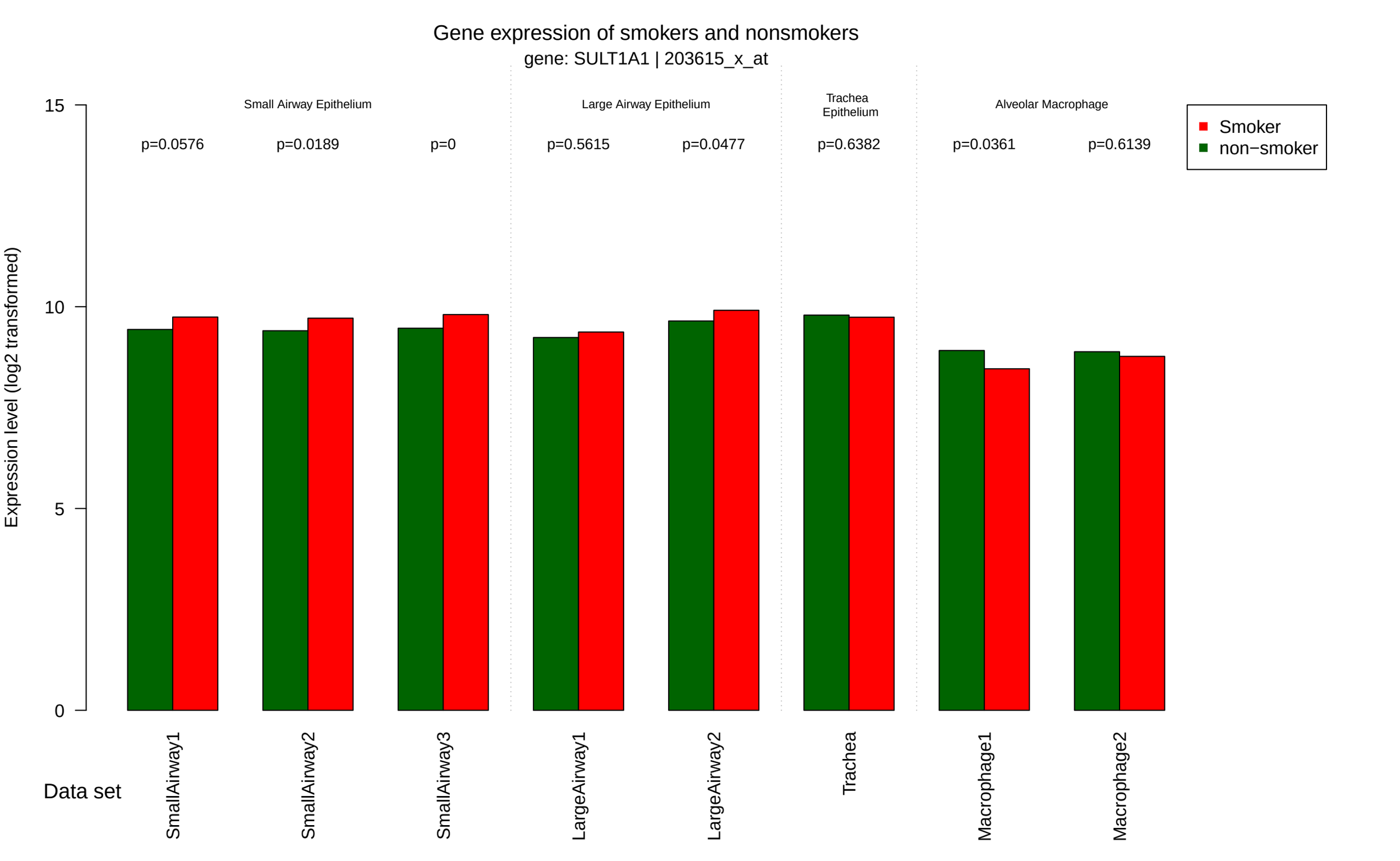
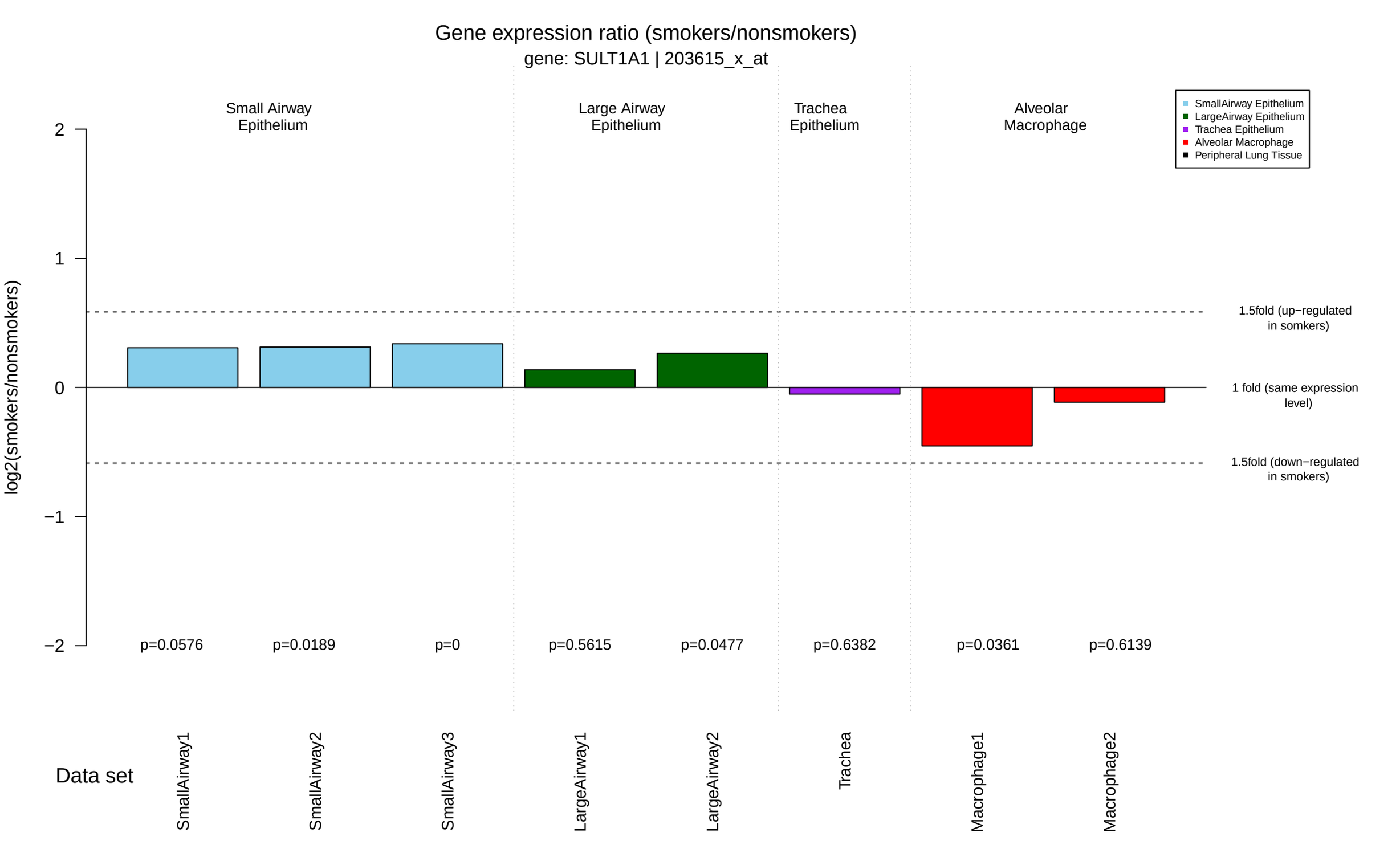
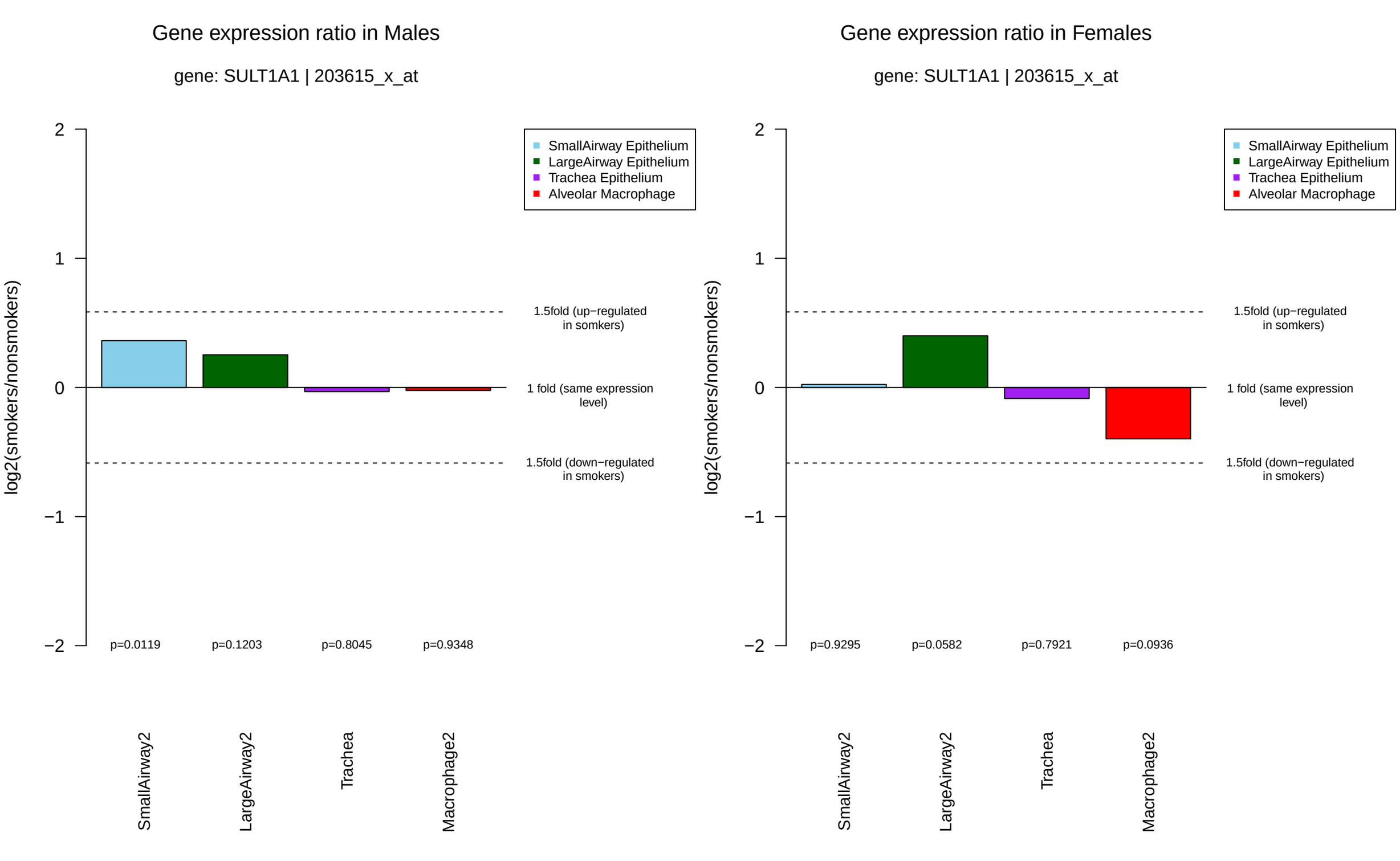
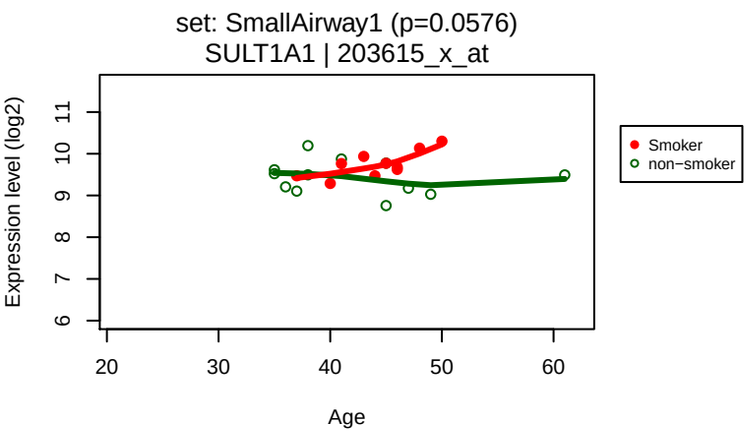
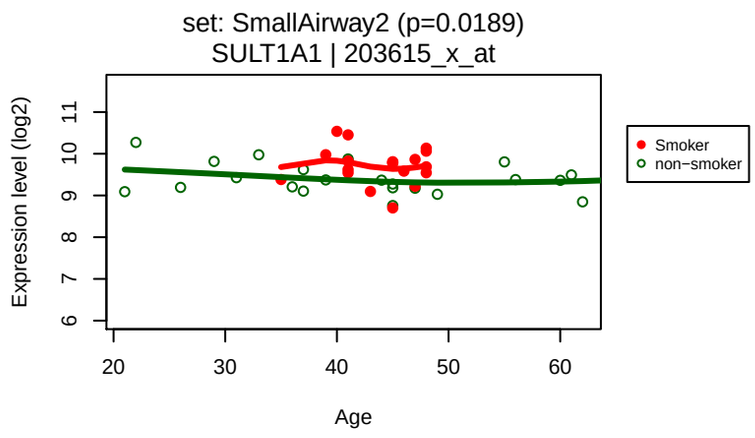
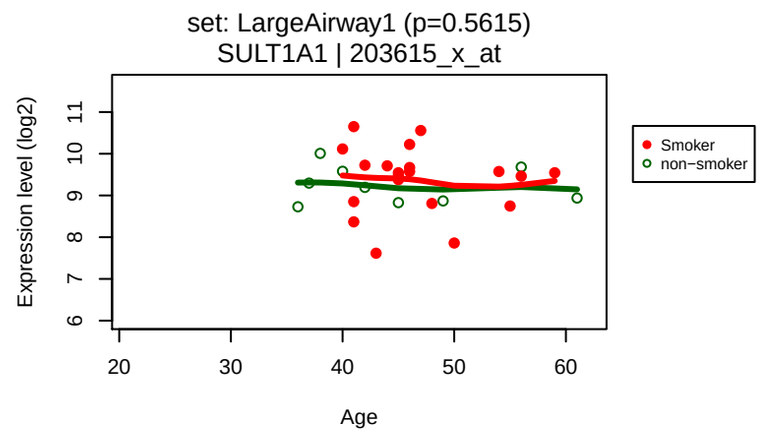

Gene expression of smokers and nonsmokers
gene: SULT1A1 | 203615_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0576
p=0.0189
p=0
p=0.5615
p=0.0477
p=0.6382
p=0.0361
p=0.6139
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

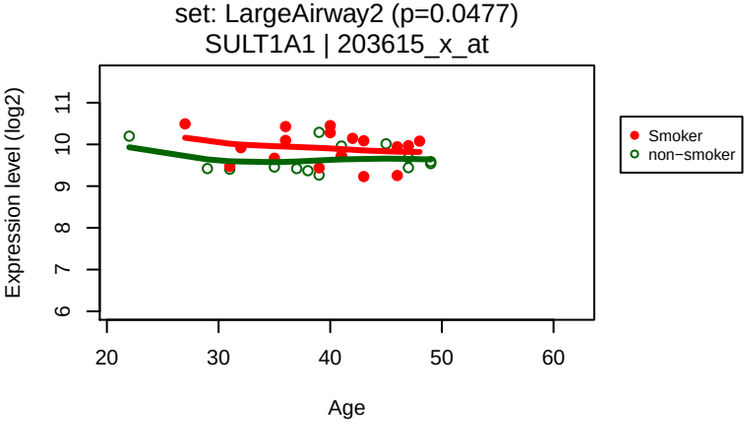

Gene expression ratio (smokers/nonsmokers)
gene: SULT1A1 | 203615_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0576
p=0.0189
p=0
p=0.5615
p=0.0477
p=0.6382
p=0.0361
p=0.6139
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

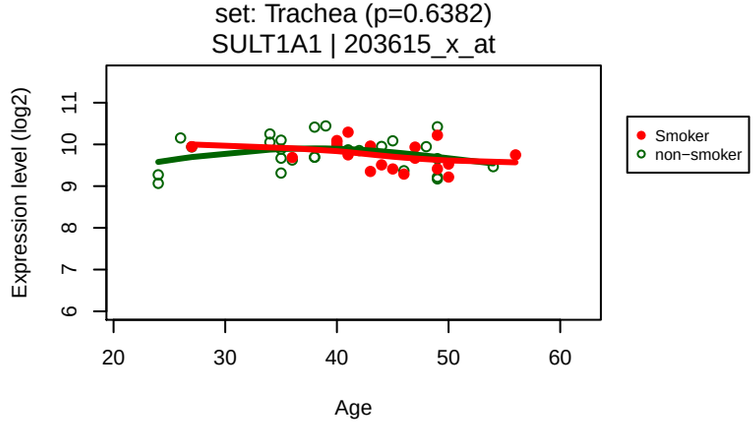

Gene expression ratio in Males
gene: SULT1A1 | 203615_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0119
p=0.1203
p=0.8045
p=0.9348
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SULT1A1 | 203615_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9295
p=0.0582
p=0.7921
p=0.0936
SmallAirway2
LargeAirway2
Trachea
Macrophage2

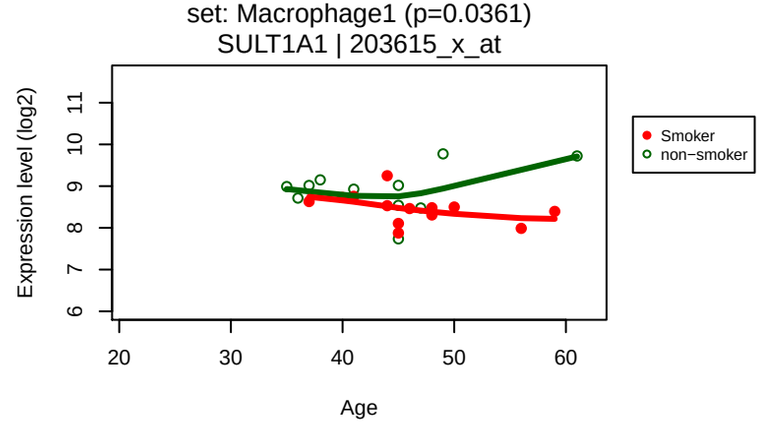

set: SmallAirway1 (p=0.0576)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker

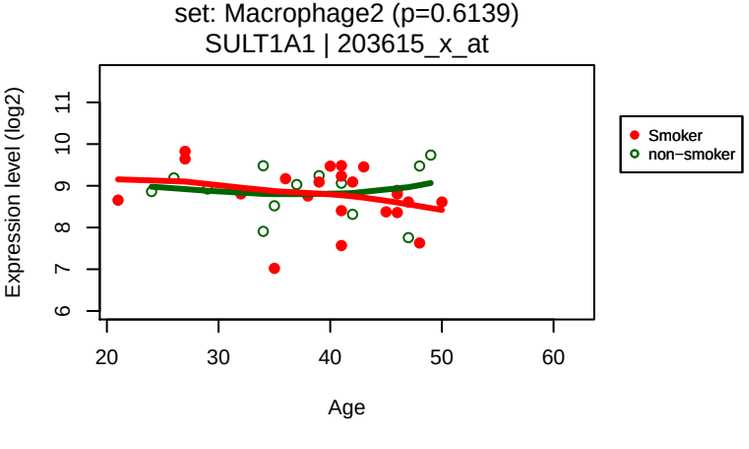

set: SmallAirway2 (p=0.0189)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker

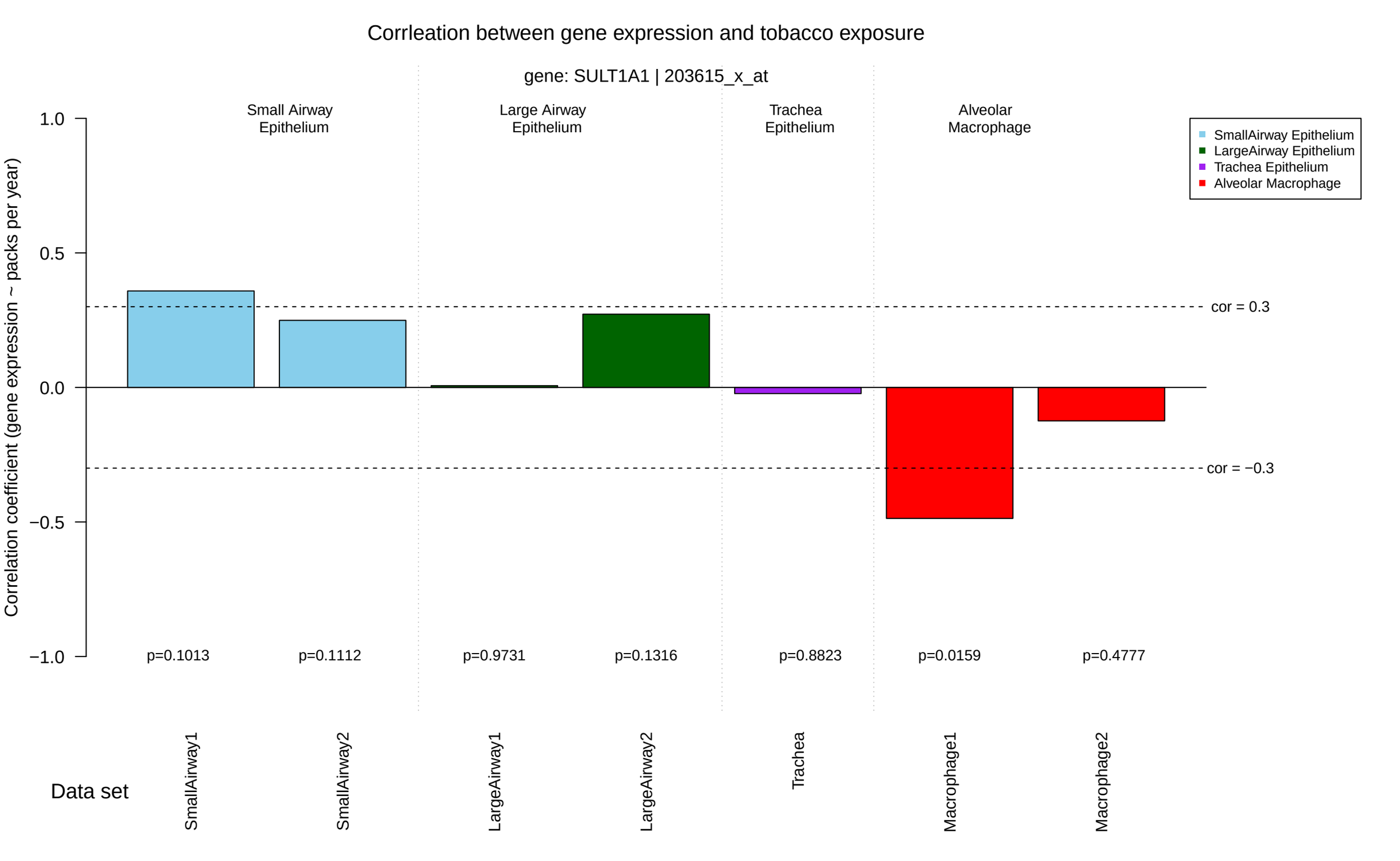

set: LargeAirway1 (p=0.5615)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0477)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6382)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0361)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6139)
SULT1A1 | 203615_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SULT1A1 | 203615_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1013
p=0.1112
p=0.9731
p=0.1316
p=0.8823
p=0.0159
p=0.4777
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2